

CCNA Security

Lab 1 - Researching Network Attacks and Security Audit Tools/Attack Tools

Objectives

Part 1: Researching Network Attacks

- Research network attacks that have occurred.
- Select a network attack and develop a short report.

Part 2: Researching Network Security Audit Tools and Attack Tools

- Research network security audit tools.
- Select a tool and develop a short report.

Background / Scenario

Attackers have developed many tools over the years to attack and compromise networks. These attacks take many forms, but in most cases, they seek to obtain sensitive information, destroy resources, or deny legitimate users access to resources. When network resources are inaccessible, worker productivity can suffer, and business income may be lost.

To understand how to defend a network against attacks, an administrator must identify network vulnerabilities. Specialized security audit software, developed by equipment and software manufacturers, can be used to help identify potential weaknesses. These same tools used by individuals to attack networks can also be used by network professionals to test the ability of a network to mitigate an attack. After the vulnerabilities are discovered, steps can be taken to help protect the network.

This lab provides a structured research project that is divided into two parts: Researching Network Attacks and Researching Security Audit Tools. Inform your instructor about which network attack(s) and network security audit tool(s) you have chosen to research. This will ensure that a variety of network attacks and vulnerability tools are reported on by the members of the class.

In Part 1, research network attacks that have actually occurred. Select one of these attacks and describe how the attack was perpetrated and the extent of the network outage or damage. Next, investigate how the attack could have been mitigated, or what mitigation techniques might have been implemented to prevent future attacks. Finally, prepare a report based on the form included in this lab.

In Part 2, research network security audit tools and attack tools. Investigate one that can be used to identify host or network device vulnerabilities. Create a one-page summary of the tool based on the form included within this lab. Prepare a short (5–10 minute) presentation to give to the class.

You may work in teams of two, with one person reporting on the network attack and the other reporting on the tools. All team members deliver a short overview of their findings. You can use live demonstrations or PowerPoint, to summarize your findings.

Required Resources

- Computer with Internet access for research
- Presentation computer with PowerPoint or other presentation software installed
- Video projector and screen for demonstrations and presentations

Part 1: Researching Network Attacks

In Part 1 of this lab, you will research real network attacks and select one on which to report. Fill in the form below based on your findings.

Step 1: Research various network attacks.

List some of the attacks you identified in your search.

Possible examples include: Code Red, Flame, Nimba, Back Orifice, Blaster, MyDoom, SQL Slammer, SMURF, Tribe flood network (TFN), Stacheldraht, Sobig, Netsky, Witty, Stuxnet and Storm.

- SQL Injection
- Malicious Code Injection
- Watering Hole Attack
- Insider Threats
- Distributed Denial of Service (DDoS) attacks
- Increased Privileges
- Access of Illegitimate Users
- Man-in-the-middle / Eavesdropping Attacks
- Zero-day attack
- Cross-Site Scripting
- Wi-Fi Eavesdropping
- DNS Spoofing

Step 2: Fill in the following form for the network attack selected.

Name of attack:	Watering Hole Attack
Type of attack:	Phishing
Dates of attacks:	2014
Computers / Organizations affected:	Forbes Web site
How it works and what it did:	
Forbes was the victim of a watering hole attack launched by a Chinese hacking group. The campaign exploited zero-day vulnerabilities in Internet Explorer and Adobe Flash Player to show malicious versions of Forbes' "Thought of the Day" feature, which loaded through a Flash widget whenever a visitor accessed a page on the site. This enabled the watering hole attack to infect any vulnerable devices that visited the Forbes website.	

Mitigation options:

Regular security testing: Organizations need to regularly test security solutions to verify that they provide the necessary defense level. This ensures that users always browse the internet securely, prevents intentional and unintentional downloads of malware or rootkits, and stops users from accessing infected or malicious websites.

Advanced threat protection: Security solutions that protect organizations against advanced attack vectors are crucial to preventing watering hole attacks. Advanced threat protection tools include behavioral analysis solutions, which give organizations a better chance of detecting zero-day exploits before attackers can target users.

System and software updates: An essential best practice for avoiding watering hole attacks is to update systems and software and install operating system patches as soon as they are made available by vendors. Attackers infect websites by discovering vulnerabilities in their code, making it imperative to spot flaws or gaps in software before cyber criminals find them.

References and info links:

<https://www.fortinet.com/resources/cyberglossary/watering-hole-attack>

Part 2: Researching Network Security Audit Tools and Attack Tools

In Part 2 of this lab, research network security audit tools and attack tools. Investigate one that can be used to identify host or network device vulnerabilities. Fill in the report below based on your findings.

Step 1: Research various network security audit tools and attack tools.

List some of the tools that you identified in your search.

Possible examples include: Microsoft Baseline Security Analyzer (MBSA), NMAP, Cisco IOS AutoSecure.. Sourceforge Network Security Analysis Tool (NSAT), Solarwinds Engineering Toolset.

Attacker tools may also be investigated, including L0phtcrack, Cain and Abel, John the Ripper, Netcat, THC Hydra, Chkrootkit, DSNIFF, Nessus, AirSnort, AirCrack, WEPCrack.

Network Security Audit Tools:

- Nessus

- Nmap
- Nexpose
- QualysGuard
- Wireshark

Attack Tools:

- Metasploit Framework
- Burp Suite
- SQLMap
- Aircrack-ng
- THC-Hydra

Step 2: Fill in the following form for the network security audit tool/attack tool selected.

Name of tool:	Wireshark
Developer:	The Wireshark team
Type of tool (character-based or GUI):	QT widget toolkit
Used on (network device or computer host):	Network device
Cost:	Free
Description of key features and capabilities of product or tool:	
Wireshark lets the user put network interface controllers that support promiscuous mode into that mode, so they can see all traffic visible on that interface, not just traffic addressed to one of the interface's configured addresses and broadcast/multicast traffic	
References and info links:	
https://en.wikipedia.org/wiki/Wireshark	

Reflection

1. What is the impact of network attacks on the operation of an organization? What are some key steps organizations can take to help protect their networks and resources?

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2. Have you actually worked for an organization or know of one where the network was compromised? If so, what was the impact on the organization and what did it do about it?

3. What steps can you take to protect your own PC or laptop computer?
